

MAT1001 Calculus for Engineers

Change of Variables and Jacobian



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Change of Variable – Single Variable Functions

Recall

$$\int_a^b f(x) dx$$

The substitution $x = g(u)$ leads to

$$\int_a^b f(x) dx = \int_\alpha^\beta f(g(u)) |g'(u)| du$$

The expression $g'(u)$ is called the **Jacobian** of the coordinate transformation $x = g(u)$

► **Example** Consider the integral

$$\int_0^2 x \cos(x^2) dx$$

Use the substitution

$$x = g(u) = \sqrt{u}$$

This substitution send the interval $[0, 2]$ onto the interval $[0, 4]$. Since

$$g'(u) = \frac{1}{2\sqrt{u}}$$

the integral becomes

$$\int_0^4 \sqrt{u} \cos(u) \frac{1}{2\sqrt{u}} du = \frac{1}{2} \int_0^4 \cos(u) du = \frac{1}{2} \sin 4$$

We want to perform similar substitutions for double integrals. Before that we will consider consider the concepts of transformations and Jacobians in two variable.

Coordinate Transformations - Recap

Now we will consider parametric equations of the form

$$x = x(u, v), \quad y = y(u, v) \quad (2)$$

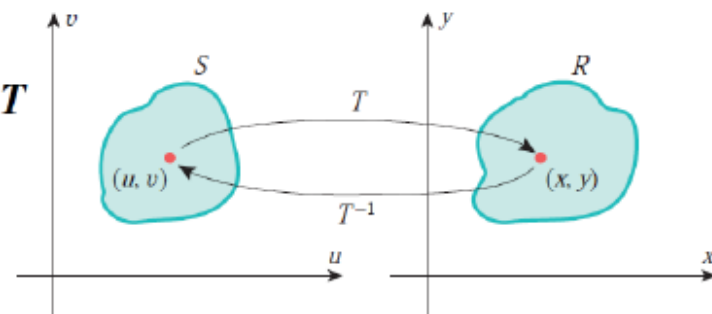
Consider

$$T(u, v) = (x(u, v), y(u, v))$$

We call T a **transformation** from the uv -plane to the xy -plane and (x, y) the **image** of (u, v) under the transformation T . We also say that T **maps** (u, v) into (x, y) . The set R of all images in the xy -plane of a set S in the uv -plane is called the **image of S under T** . If distinct points in the uv -plane have distinct images in the xy -plane, then T is said to be **one-to-one**. In this case the equations in (2) define u and v as functions of x and y , say

$$u = u(x, y), \quad v = v(x, y)$$

This transformation is denoted by T^{-1} and is called the **inverse of T**



Example

► **Example** Let T be the transformation from the uv -plane to the xy -plane defined by the equations

$$x = \frac{1}{4}(u + v), \quad y = \frac{1}{2}(u - v) \quad (3)$$

- (a) Find $T(1, 3)$.
- (b) Sketch the constant v -curves corresponding to $v = -2, -1, 0, 1, 2$.
- (c) Sketch the constant u -curves corresponding to $u = -2, -1, 0, 1, 2$.
- (d) Sketch the image under T of the square region in the uv -plane bounded by the lines $u = -2, u = 2, v = -2$, and $v = 2$.

Solution (a). Substituting $u = 1$ and $v = 3$ in (3) yields $T(1, 3) = (1, -1)$.

Solutions (b and c). In these parts it will be convenient to express the transformation equations with u and v as functions of x and y . From (3)

$$4x = u + v, \quad 2y = u - v$$

Combining these equations gives

$$4x + 2y = 2u, \quad 4x - 2y = 2v$$

or

$$2x + y = u, \quad 2x - y = v$$

Example

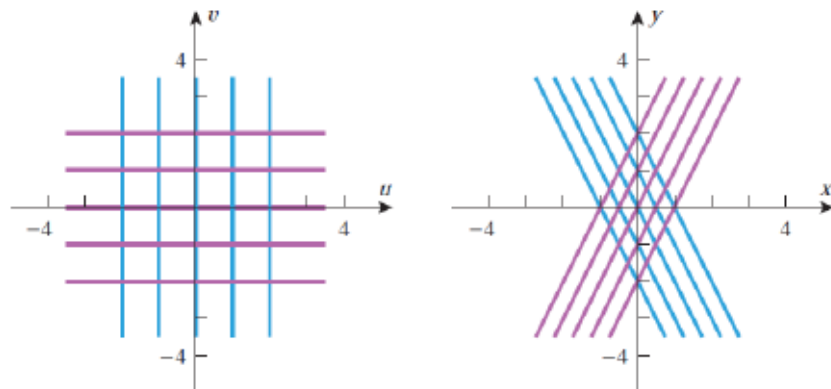
Thus, the constant v -curves corresponding to $v = -2, -1, 0, 1$, and 2 are

$$2x - y = -2, \quad 2x - y = -1, \quad 2x - y = 0, \quad 2x - y = 1, \quad 2x - y = 2$$

and the constant u -curves corresponding to $u = -2, -1, 0, 1$, and 2 are

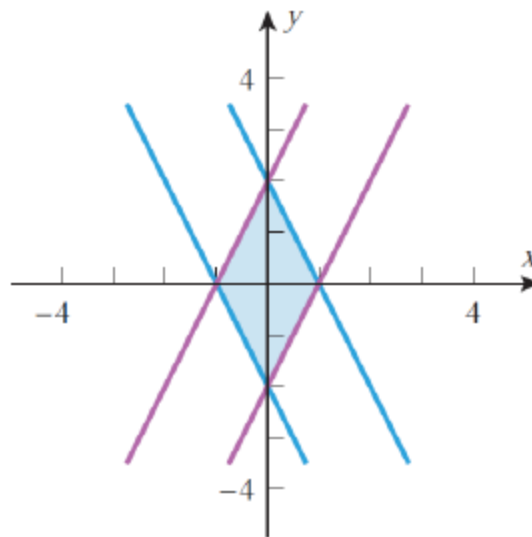
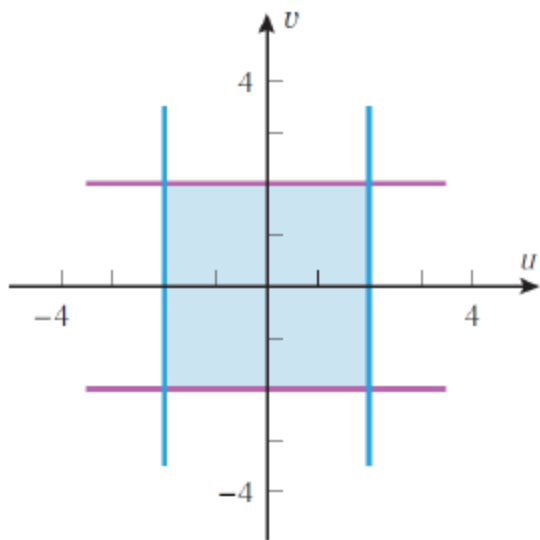
$$2x + y = -2, \quad 2x + y = -1, \quad 2x + y = 0, \quad 2x + y = 1, \quad 2x + y = 2$$

the constant v -curves are shown in purple and the constant u -curves in blue.



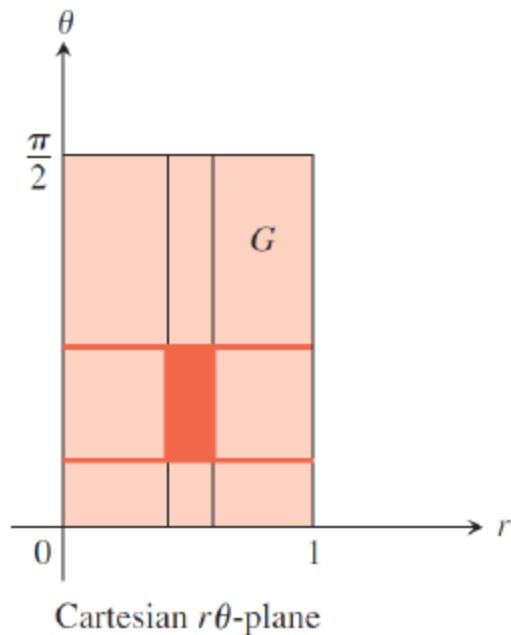
Example

Solution (d). The image of a region can often be found by finding the image of its boundary. In this case the images of the boundary lines $u = -2$, $u = 2$, $v = -2$, and $v = 2$ enclose the diamond-shaped region in the xy -plane

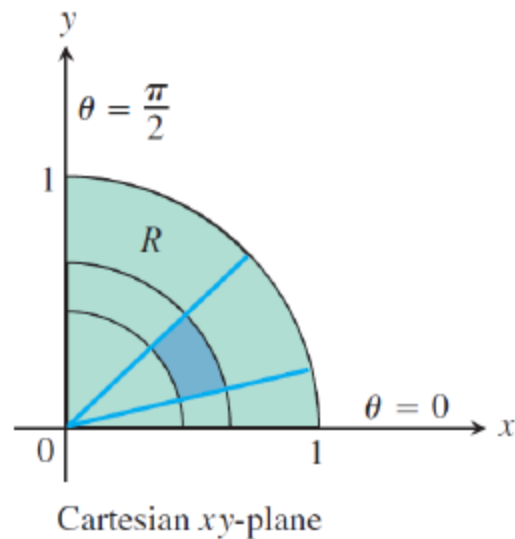


Example

- **Example** Figure shows how the polar coordinate transformation $x = r \cos \theta$, $y = r \sin \theta$ transform the rectangle $G: 0 \leq r \leq 1, 0 \leq \theta \leq \pi/2$, into the Cartesian quarter circle R bounded by $x^2 + y^2 = 1$ in the first quadrant of the xy -plane.



$\xrightarrow{\text{pink arrow}}$
 $x = r \cos \theta$
 $y = r \sin \theta$



Jacobian – Two Variable Functions - Recap

DEFINITION If T is the transformation from the uv -plane to the xy -plane defined by the equations $x = x(u, v)$, $y = y(u, v)$, then the **Jacobian of T** is denoted by $J(u, v)$ or by $\partial(x, y)/\partial(u, v)$ and is defined by

$$J(u, v) = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \frac{\partial x}{\partial u} \frac{\partial y}{\partial v} - \frac{\partial y}{\partial u} \frac{\partial x}{\partial v}$$

► **Example** Find the Jacobian for the polar coordinate transformation $x = r \cos \theta$, $y = r \sin \theta$,

Solution For polar coordinates, we have r and θ in place of u and v . With $x = r \cos \theta$ and $y = r \sin \theta$, the Jacobian is

$$J(r, \theta) = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r(\cos^2 \theta + \sin^2 \theta) = r.$$

Change of Variables in Double Integration

CHANGE OF VARIABLES FORMULA FOR DOUBLE INTEGRALS

If the transformation $x = x(u, v)$, $y = y(u, v)$ maps the region S in the uv -plane into the region R in the xy -plane, and if the Jacobian $\partial(x, y)/\partial(u, v)$ is nonzero and does not change sign on S , then with appropriate restrictions on the transformation and the regions it follows that

$$\iint_R f(x, y) dA_{xy} = \iint_S f(x(u, v), y(u, v)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| dA_{uv}$$

where we have attached subscripts to the dA 's to help identify the associated variables.

► **Example** write the Cartesian integral $\iint_R f(x, y) dx dy$ as a polar integral.

Since we assume $r \geq 0$ when integrating in polar coordinates, $|J(r, \theta)| = |r| = r$, so that

$$\iint_R f(x, y) dx dy = \iint_G f(r \cos \theta, r \sin \theta) \left| \frac{\partial(x, y)}{\partial(r, \theta)} \right| dr d\theta = \iint_G f(r \cos \theta, r \sin \theta) r dr d\theta.$$

Example

► **Example** Evaluate

$$\iint_R e^{xy} dA$$

where R is the region enclosed by the lines $y = \frac{1}{2}x$ and $y = x$ and the hyperbolas $y = 1/x$ and $y = 2/x$

Solution. look for a transformation in which the boundary curves in the xy -plane become constant v -curves and constant u -curves.

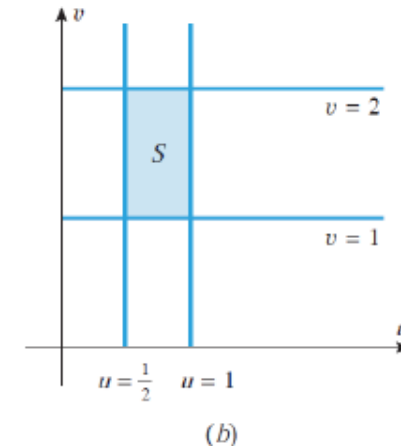
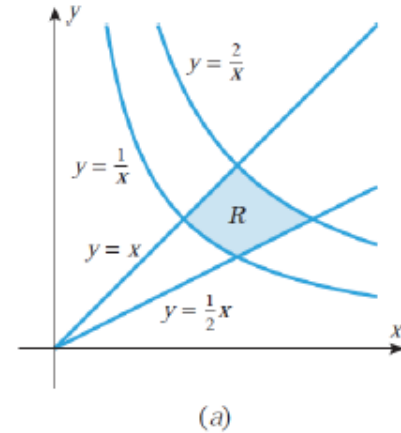
$$\frac{y}{x} = \frac{1}{2}, \quad \frac{y}{x} = 1, \quad xy = 1, \quad xy = 2$$

which suggests the transformation

$$u = \frac{y}{x}, \quad v = xy \quad (11)$$

With this transformation the boundary curves in the xy -plane are constant u -curves and constant v -curves corresponding to the lines

$$u = \frac{1}{2}, \quad u = 1, \quad v = 1, \quad v = 2$$



Example

This yields

$$x = \sqrt{v/u}, \quad y = \sqrt{uv}$$

from which we obtain

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} -\frac{1}{2u}\sqrt{\frac{v}{u}} & \frac{1}{2\sqrt{uv}} \\ \frac{1}{2}\sqrt{\frac{v}{u}} & \frac{1}{2}\sqrt{\frac{u}{v}} \end{vmatrix} = -\frac{1}{4u} - \frac{1}{4u} = -\frac{1}{2u}$$

Thus, from Formula (9), but with the notation dA rather than dA_{xy} ,

$$\begin{aligned} \iint_R e^{xy} dA &= \iint_S e^v \left| -\frac{1}{2u} \right| dA_{uv} = \frac{1}{2} \iint_S \frac{1}{u} e^v dA_{uv} \\ &= \frac{1}{2} \int_1^2 \int_{1/2}^1 \frac{1}{u} e^v du dv = \frac{1}{2} \int_1^2 e^v \ln |u| \Big|_{u=1/2}^1 dv \\ &= \frac{1}{2} \ln 2 \int_1^2 e^v dv = \frac{1}{2} (e^2 - e) \ln 2 \end{aligned}$$

DEFINITION If T is the transformation from uvw -space to xyz -space defined by the equations $x = x(u, v, w)$, $y = y(u, v, w)$, $z = z(u, v, w)$, then the **Jacobian of T** is denoted by $J(u, v, w)$ or $\partial(x, y, z)/\partial(u, v, w)$ and is defined by

$$J(u, v, w) = \frac{\partial(x, y, z)}{\partial(u, v, w)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix}$$

Change of Variables in Triple Integration

CHANGE OF VARIABLES FORMULA FOR TRIPLE INTEGRALS

If the transformation $x = x(u, v, w)$, $y = y(u, v, w)$, $z = z(u, v, w)$ maps the region S in uvw -space into the region R in xyz -space, and if the Jacobian $\partial(x, y, z)/\partial(u, v, w)$ is nonzero and does not change sign on S , then with appropriate restrictions on the transformation and the regions it follows that

$$\iiint_R f(x, y, z) dV_{xyz} = \iiint_S f(x(u, v, w), y(u, v, w), z(u, v, w)) \left| \frac{\partial(x, y, z)}{\partial(u, v, w)} \right| dV_{uvw} \quad (14)$$

Example

► **Example** Find the volume of the region G enclosed by the ellipsoid

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

Solution. The volume V is given by the triple integral

$$V = \iiint_G dV$$

To evaluate this integral, we make the change of variables

$$x = au, \quad y = bv, \quad z = cw \quad (15)$$

which maps the region S in uvw -space enclosed by a sphere of radius 1 into the region G in xyz -space. This can be seen from (15) by noting that

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \quad \text{becomes} \quad u^2 + v^2 + w^2 = 1$$

Example

The Jacobian of (15) is

$$\frac{\partial(x, y, z)}{\partial(u, v, w)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix} = \begin{vmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{vmatrix} = abc$$

Thus, from Formula (14), but with the notation dV rather than dV_{xyz} ,

$$V = \iiint_G dV = \iiint_S \left| \frac{\partial(x, y, z)}{\partial(u, v, w)} \right| dV_{uvw} = abc \iiint_S dV_{uvw}$$

The last integral is the volume enclosed by a sphere of radius 1, which we know to be $\frac{4}{3}\pi$.

Thus, the volume enclosed by the ellipsoid is $V = \frac{4}{3}\pi abc$.

Example

► **Example** Jacobian of the transformation $x = r \cos \theta$, $y = r \sin \theta$, $z = z$ is

$$\frac{\partial(x, y, z)}{\partial(r, \theta, z)} = r$$

Jacobian of the transformation $x = \rho \sin \phi \cos \theta$, $y = \rho \sin \phi \sin \theta$, $z = \rho \cos \phi$ is

$$\frac{\partial(x, y, z)}{\partial(\rho, \phi, \theta)} = \rho^2 \sin \phi$$

Thus,

$$\iiint_G f(x, y, z) dV = \iiint_{\text{appropriate limits}} f(r \cos \theta, r \sin \theta, z) \frac{\partial(x, y, z)}{\partial(r, \theta, z)} dz dr d\theta$$

$$\iiint_G f(x, y, z) dV = \iiint_{\text{appropriate limits}} f(\rho \sin \phi \cos \theta, \rho \sin \phi \sin \theta, \rho \cos \phi) \frac{\partial(x, y, z)}{\partial(\rho, \phi, \theta)} d\rho d\phi d\theta$$



Thank You...

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